

SOFTWARE REQUIREMENT SPECIFICATION & BUSINESS PROPOSAL

FreightFlow ERP

Integrated Freight Forwarding Enterprise Resource Planning System

Import	Export	Customs	Container	Transport	Billing	Analytics
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01 Executive Summary

FreightFlow ERP is a purpose-built, cloud-native Enterprise Resource Planning solution engineered specifically for freight forwarding and international logistics companies. This document defines the complete functional scope, technical architecture, and delivery strategy for a fully digitised logistics operations platform.

The global freight forwarding industry is undergoing rapid digital transformation driven by increasing demands for real-time shipment visibility, regulatory compliance, and end-to-end operational efficiency. Legacy paper-based and semi-automated processes are no longer competitive. FreightFlow ERP addresses these challenges by consolidating the entire freight lifecycle — from the first customer inquiry through to the final invoice — into a single integrated platform.

Key Business Drivers

- Eliminate manual, paper-based documentation across all import and export workflows
- Achieve real-time visibility into shipment status, container locations, and vessel ETAs
- Automate customs clearance documentation (BOE, Shipping Bill) and duty calculation
- Centralise CRM, quotation, billing, and accounts receivable into one unified system
- Enable data-driven decision making through live dashboards, analytics, and AI-powered insights
- Ensure full GST compliance and audit-ready financial reporting at all times

Solution at a Glance

Capability	Description
10 Core ERP Modules	CRM, Quotation, Shipment Jobs, Documents, Booking, Container, Customs, Billing, Notifications, Analytics
Dual Workflow Support	Complete Import and Export shipment lifecycle management from inquiry to job closure
Cloud-Native Platform	AWS / Azure hosted with auto-scaling, high availability, and 99.9% SLA uptime
Mobile-Ready Application	iOS and Android mobile app for field operations, transport staff, and management
AI-Enhanced Features (Phase 3)	OCR document reading, delay prediction, smart quotations, and AI customer chatbot
Compliance-First Design	GST-compliant billing, IEC / BOE compliance, and immutable audit trail on all records
API-First Architecture	Integrations with shipping lines, customs portals, WhatsApp, and payment gateways

STRATEGIC RECOMMENDATION

A phased delivery approach is strongly recommended — delivering a fully functional Minimum Viable Product (MVP) as the first major milestone to generate early business value, followed by progressive module rollout toward the complete enterprise platform. This approach reduces delivery risk and ensures the platform evolves based on real operational feedback.

02 Project Overview

2.1 Project Background

Freight forwarding companies operate across a complex ecosystem of shipping lines, customs authorities, transport vendors, CFS operators, port terminals, and end customers. The typical operational model involves multiple disconnected tools — Excel sheets for rate management, lengthy email chains for document exchange, separate accounting software for billing, and manual customs filing processes — resulting in data silos, human errors, costly delays, and a consistently poor customer experience.

FreightFlow ERP replaces this fragmented operational model with a unified, role-aware platform that tracks every shipment job from inquiry through to closure, maintains a complete and immutable audit trail, and provides every stakeholder — from field operations staff to senior management — with accurate, real-time information precisely when they need it.

2.2 Project Objectives

Objective	Target Outcome
Digitise Import & Export workflows end-to-end	Zero paper-based job processing within six months of go-live
Reduce customer shipment status inquiries	80% reduction in inbound status calls via customer self-service portal
Accelerate invoice generation	Invoice dispatched within 24 hours of job closure (versus 5–7 days manual)
Improve document compliance	100% document checklist adherence enforced before customs filing
Enable real-time container tracking	Live container status accessible to operations, finance, and customers
Provide revenue visibility	Daily P&L per shipment job; live revenue dashboards for management
Improve customs clearance accuracy	Eliminate rejected BOE and Shipping Bill submissions due to documentation errors

2.3 Scope of the ERP System

The scope of FreightFlow ERP encompasses the following operational domains:

- Customer Relationship Management (CRM) and freight inquiry management
- Freight quotation preparation, approval workflow, and rate management
- Import and Export shipment job lifecycle management
- Document Management System (DMS) with verification workflows
- Shipping line booking management and vessel schedule tracking
- Container allocation and transportation coordination
- Customs clearance — Bill of Entry (Import) and Shipping Bill (Export)
- Billing, invoicing, GST compliance, and accounts management
- Automated notification engine — WhatsApp, Email, and SMS
- Dashboards, management reports, and business analytics
- Mobile application for field and management users
- Third-party API integrations with customs portals, shipping lines, and payment gateways

03 Business Objectives

3.1 Strategic Objectives

#	Objective	Business Impact
1	Operational Digitisation	Eliminate manual data entry errors; reduce per-job processing time by 60%
2	Customer Experience Enhancement	Real-time shipment updates; self-service portal for importers and exporters
3	Compliance & Risk Management	Automated GST and customs compliance; immutable audit trail across all records
4	Revenue Optimisation	Accurate capture of all charges per job; systematic elimination of revenue leakage
5	Business Scalability	Support 10× volume growth without a proportional increase in headcount
6	Data-Driven Intelligence	AI-powered insights for pricing, delay prediction, and demand forecasting

3.2 Operational Pain Points Addressed

Current Pain Point	FreightFlow ERP Solution
Manual quotation preparation takes 2-3 hours per quote	Automated rate cards; professional PDF quotation generated in minutes
No single centralised shipment tracking system	Real-time job dashboard with status timeline visible to all stakeholders
Document management via email and WhatsApp threads	Structured DMS with version control, checklists, and expiry alerts
Frequent manual errors in BOE and Shipping Bill preparation	Guided customs module with mandatory pre-filing validation checks
Invoice generation delayed by 5-7 days after job closure	Automated invoice triggered upon job closure; GST-compliant PDF dispatched same day
No visibility into outstanding customer receivables	Live AR dashboard with full aging analysis and payment reminders
Transport and driver coordination conducted entirely by phone	In-app transport allocation, trip management, and real-time driver GPS tracking
No automated customer communication at shipment milestones	Automated WhatsApp and email notifications triggered at every key milestone

04 User Roles & Permissions

4.1 User Roles & Permissions Matrix

Role	Module Access	Permission Level
Super Admin	All Modules	Full CRUD access plus system-wide configuration
Branch Manager	All operational modules and reports	Full CRUD restricted to assigned branch

Role	Module Access	Permission Level
Operations Executive	Jobs, Documents, Booking, Transport	Create, Edit, and View
Customs Executive	Customs Module and Documents	Create, Edit, and View
Accounts Executive	Billing, Payments, and GST Reports	Create, Edit, and View
Sales Executive	CRM and Quotation	Create, Edit, and View
Transport Coordinator	Transport and Container Module	Create, Edit, and View
Customer — Portal User	Own shipments, documents, and invoices	View Only — restricted to own records
Auditor	All Modules (read-only access)	View Only with export capability
IT Admin	System Config and User Management	Administrative configuration only

05 ERP Modules — Detailed Specifications

MODULE 1 | CUSTOMER RELATIONSHIP MANAGEMENT (CRM)

The CRM module is the foundational master data registry for all customer interactions. Every shipment job, quotation, and invoice is linked to a customer record maintained within this module. It serves as the single source of truth for all customer-facing data.

Customer Master — Field Specifications

Field	Data Type	Description
Customer ID	Auto-generated UUID	Unique alphanumeric customer identifier; system-assigned on registration
Customer Name	Text	Legal registered name of the business entity
Customer Type	Dropdown	Importer / Exporter / Both / Freight Agent
GST Number	Text — 15 characters	GSTIN with format validation; mandatory for all Indian business entities
IEC Code	Text — 10 characters	Importer Exporter Code; mandatory for all international trade transactions
PAN Number	Text	Permanent Account Number; used for financial and tax compliance
Registered Address	Multi-field	Full address with city, state, PIN code, and country
Contact Persons	Repeatable Group	Name, mobile number, email address, and designation for each contact
Credit Limit	Currency	Approved credit ceiling for accounts receivable management
Payment Terms	Dropdown	Cash / 15 Days / 30 Days / 45 Days — used in invoice due date calculation
KYC Documents	File Upload	GST certificate, IEC copy, and PAN card stored securely in DMS
Customer Status	Toggle	Active / Inactive / Blacklisted — controls access to new job creation

Customer Analytics & History

- Complete shipment history per customer — Import volume, Export volume, and combined totals
- Revenue generated per customer viewable monthly, quarterly, and annually
- Full payment history with outstanding invoice aging and credit utilisation tracking
- Quotation conversion rate — total quotes sent versus jobs successfully created
- Top commodities, trade lanes, and preferred shipping lines per customer

MODULE 2 | INQUIRY & QUOTATION MANAGEMENT

This module captures all inbound customer freight inquiries, enables the commercial team to prepare accurate and competitive quotations, generates professional PDF proposals, and manages the full approval workflow through to job creation.

Inquiry Form — Key Fields

Field	Description
Inquiry Number	Auto-generated reference number in format INQ-YYYY-NNNN
Inquiry Date	Date on which the customer inquiry was received
Customer	Linked directly from the CRM customer master
Shipment Type	Import or Export
Mode of Transport	Ocean FCL / Ocean LCL / Air Freight / Road Transport
Origin & Destination	Port code, airport, or city for both origin and destination
Commodity	Description and nature of the cargo to be shipped
Container Type	20GP / 40GP / 40HC / LCL cargo / Air ULD
Weight & Volume	Gross weight in KG, volume in CBM, and total package count
Expected Shipment Date	Target shipment or arrival date as requested by the customer
Special Requirements	Hazardous material class, temperature control range, or Out-of-Gauge dimensions
Inquiry Status	New / Quote Sent / Converted to Job / Lost — updated at each stage

Quotation Builder — Key Features

- Multi-line charge builder covering Ocean Freight, THC, BL Fee, CFS, Documentation, Customs, Transport, and Handling charges
- Rate validity period management with automatic expiry notification dispatched to the sales team
- Container-type-based rate calculation — configurable per TEU, per CBM, or per KG
- Multi-currency support with exchange rate management and rate-locking at quotation date
- Automatic PDF quotation generation with company letterhead, terms and conditions, and validity date
- Email dispatch of quotation PDF directly from the system with delivery status tracking
- Approval workflow — Sales Executive submits; Branch Manager approves for quotes above defined threshold
- Quotation version control (v1, v2, v3) with side-by-side version comparison capability

MODULE 3 | SHIPMENT JOB MANAGEMENT

The Job Management module is the central operational hub of FreightFlow ERP. Every shipment — import or export — is managed as a numbered Job with a unique reference, a complete status timeline, and a full audit trail from creation through to final closure.

Job Master — Key Fields

Field	Description
Job Number	Auto-generated — IMP-2026-0001 format for Import; EXP-2026-0001 for Export
Job Type	Import or Export
Linked Quotation	Reference to the approved quotation; auto-populates job fields
Customer	Linked from CRM; contact details auto-populated on selection
Consignor & Consignee	Full shipper and receiver details including address and contact
Trade Lane	Origin Country → Destination Country
Port of Loading (POL)	Origin port — UNLOCODE and full name
Port of Discharge (POD)	Destination port — UNLOCODE and full name
Final Destination	Inland point of delivery if different from port of discharge
Incoterms	EXW / FOB / CIF / CFR / DAP / DDP — governs risk and cost split
Cargo Details	Commodity, HS Code, gross weight, volume in CBM, and package count
Container Details	Type, quantity, and container numbers once allocated by shipping line
Special Handling	Hazmat IMDG class, temperature range, or OOG dimensions
Job Status	Active / On Hold / Customs Cleared / Delivered / Closed
Handling Branch	Company branch responsible for managing this shipment

Shipment Status Timeline — Export & Import

S t a g e	Export Jobs	Import Jobs
	01 Customer Inquiry / Quotation Raised	Shipment Pre-Alert Received from Shipper or Agent
	02 Booking Confirmed with Shipping Line	ETA Confirmed — Vessel Voyage Details Recorded
	03 Empty Container Allocated	Document Collection Initiated

S t a g e	Export Jobs	Import Jobs
	04 Factory Stuffing Completed	Bill of Entry (BOE) Filed on ICEGATE
	05 Customs Let Export Order (LEO) Obtained	Import Duty Calculated and Payment Confirmed
	06 Port Gate-In Confirmed	Customs Out of Charge (OOC) Granted
	07 Vessel Departed — ATD Recorded	Delivery Order (DO) Collected from Shipping Line
	08 Bill of Lading (BL) Issued	Container Delivered to Consignee Premises
	09 Invoice Generated and Dispatched	Empty Container Returned to Shipping Line Depot
	10 Job Closed	Invoice Generated — Job Officially Closed

MODULE 4 | DOCUMENT MANAGEMENT SYSTEM (DMS)

The DMS provides a structured, role-controlled document repository for all shipment-related documents. It eliminates document loss, enforces version control and compliance checklists, and ensures that all required documents are verified before critical workflow milestones are reached.

Document Types Managed

Document	Applicable To	Mandatory For
Commercial Invoice	Both Import & Export	Customs filing and Letter of Credit negotiation
Packing List	Both Import & Export	Customs filing and container stuffing verification
Bill of Lading (BL)	Import & Export	Delivery Order release and bank negotiation
Bill of Entry (BOE)	Import	Customs assessment, clearance, and duty payment
Shipping Bill	Export	LEO grant, Drawback claim, and RODTEP benefit
Delivery Order (DO)	Import	Container and cargo release from port or CFS terminal
E-Way Bill	Domestic Road Transport	Mandatory for all road movements within India
Certificate of Origin (COO)	Export	Preferential duty claims and FTA benefit eligibility
Airway Bill (AWB)	Air Shipments	Air cargo release at destination airport
Insurance Certificate	Both Import & Export	Cargo insurance claim processing
Test or Inspection Certificate	Regulated Cargo	FSSAI, BIS, and customs examination compliance

Document	Applicable To	Mandatory For
Letter of Credit (LC)	Export	Bank negotiation and LC document presentation

Core DMS Features

- Drag-and-drop document upload with file type validation — supports PDF, JPG, PNG, and XLSX formats
- Mandatory document checklist enforced per job type — system prevents customs filing if required documents are absent
- Full version control — revised documents replace previous versions while all prior versions are retained with a change log
- Document status workflow: Pending Upload → Uploaded → Under Review → Verified or Rejected
- Rejection workflow with a mandatory reason field to guide the operations team on re-submission requirements
- Expiry tracking for COO, Insurance, and regulatory certificates — automated alerts at 30, 15, and 7 days prior to expiry
- Bulk download — generate a complete ZIP archive of all job documents with a single action
- Customer portal access — importers and exporters view and download their own documents without contacting the operations team

MODULE 5 | BOOKING & SHIPPING MANAGEMENT

This module manages the commercial booking process with shipping lines, vessel schedule tracking, ETA and ETD management, and the complete Bill of Lading lifecycle from draft review through to final issuance.

Feature	Description
Booking Request	Generate structured booking instructions to the shipping line directly from job data; eliminates manual re-entry
Booking Confirmation	Record the shipping line booking reference and BKG confirmation number against the job
Vessel & Voyage Details	Record vessel name, IMO number, voyage number, and service route for full traceability
Sailing Schedule	ETD from port of loading, ETA at port of discharge, and transit time in days
Cut-Off Dates	SI cutoff, VGM cutoff, and port gate-in cutoff dates — alerts triggered as deadlines approach
BL Type Configuration	Original BL / Sea Waybill / Express BL / Surrendered BL — configurable per job
BL Draft Review	Upload and review BL draft; approval by operations team required before confirming issuance
BL Issuance	Record final BL number, issuance date, and courier tracking reference for original BL sets
Vessel Tracking	Real-time vessel position tracking via AIS data API integration
ETA Change Alerts	Automatic customer notification triggered when ETA shifts by more than 24 hours

MODULE 6 | CONTAINER & TRANSPORT MANAGEMENT

This module coordinates the complete physical movement of containers — from empty pickup through factory stuffing and port gate-in for exports, and from port discharge through consignee delivery and depot return for imports.

Export Container Workflow

- Empty container pickup order raised — defines origin depot or yard and destination factory or warehouse
- Trailer allocation and driver assignment with mobile app GPS integration for real-time tracking
- Factory or warehouse stuffing appointment scheduling; appointment confirmed and notified to the driver
- Container seal number recorded immediately after stuffing is completed
- Port gate-in tracking — confirmed gate-in timestamp, vehicle weight, and VGM submission recorded

Import Container Workflow

- Delivery Order (DO) collection tracking from the shipping line following BL surrender or freight clearance
- Container release from port, CFS, or ICD on presentation of the DO — release event recorded in the system
- Trailer and driver allocation for delivery to the consignee premises; delivery window communicated via app
- Delivery confirmation — timestamp, GPS coordinates, and receiver signature captured via mobile application
- Empty container return to shipping line depot — empty return receipt document uploaded in DMS
- Detention and demurrage calculation with automated alerts raised at day 5 and day 8 of the free-time window

Transport Coordination — Data Captured

Field	Description
Vehicle Registration Number	Truck or trailer plate number for each trip
Driver Name & Mobile Number	Used for in-app GPS tracking and direct operational communication
Transport Vendor	Third-party transporter company name if the trip is outsourced
Trip Type	Empty Pickup / Factory Stuffing / Port Gate-In / Consignee Delivery / Empty Return
Scheduled Date & Time	Planned pickup or delivery window communicated to driver
Actual Completion Date & Time	Auto-captured on trip completion via the driver mobile application
GPS Tracking	Real-time driver location visible to transport coordinator and operations team
Transport Charges	Cost captured per trip for job P&L reconciliation and vendor billing

MODULE 7 | CUSTOMS CLEARANCE MODULE

The Customs module manages all regulatory filing activities for both import (Bill of Entry) and export (Shipping Bill) shipments, covering duty calculation, examination status management, and end-to-end clearance milestone tracking.

Import Customs — Bill of Entry (BOE)

Feature	Description
BOE Number & Filing Date	Record the BOE reference number and exact date of filing on ICEGATE
Customs Assessment Location	Customs station or port of entry where the BOE is assessed
Assessment & Duty Details	Assessable value, Basic Customs Duty (BCD), IGST, and Social Welfare Surcharge

Feature	Description
Duty Calculation Engine	Automatic duty computation based on HS Code and applicable tariff rates
Duty Payment Record	Duty paid amount, challan number, bank reference, and payment date
Examination Status	No Examination / First Check / Second Check — with officer name and date
Examination Remarks	Customs officer queries, objections raised, and resolution actions taken
Out of Charge (OOC)	OOC date recorded — confirms cargo officially released by customs authority
BOE Document Upload	Assessed BOE PDF uploaded directly into DMS and linked to the job record

Export Customs — Shipping Bill

Feature	Description
Shipping Bill Number	SB number assigned by the ICEGATE Customs EDI system upon filing
Shipping Bill Type	Free Shipping Bill / Drawback / RODTEP / Ex-Bond
Let Export Order (LEO)	LEO date recorded — confirms customs authorisation for vessel loading
Drawback & RODTEP Claims	Claim amount, ARN reference number, and refund status tracking
IGST Refund Status	IGST refund claim status for registered export transactions
Export General Manifest (EGM)	EGM number recorded after vessel departure from the Indian port
Examination Status	RMS Facilitated / First Appraisalment / Examination — with officer details
Examination Remarks	Officer observations, queries raised, and compliance responses recorded

MODULE 8 | BILLING & ACCOUNTS MODULE

The Billing module converts completed or in-progress shipment jobs into accurate, GST-compliant invoices. It manages vendor expenses, tracks customer receivables, and provides real-time financial visibility across every shipment job handled by the company.

Charge Types — Revenue & Cost

Charge	Type	Applicable To
Ocean Freight	Revenue & Cost	Both Import and Export
Air Freight	Revenue & Cost	Air Shipments Only
THC — Terminal Handling Charge	Revenue & Cost	Both Import and Export
BL / AWB Documentation Fee	Revenue	Both Import and Export
CFS Charges	Revenue & Cost	Import FCL and all LCL Shipments
Customs Examination Charges	Revenue & Cost	Both Import and Export
Transportation Charges	Revenue & Cost	Both Import and Export

Charge	Type	Applicable To
Import Duty & IGST on Import	Pass-Through Cost	Import Only
Port & Terminal Charges	Revenue & Cost	Both Import and Export
Agency Commission	Revenue	Both Import and Export
Handling Charges	Revenue	Both Import and Export
Detention & Demurrage	Revenue & Cost	Both Import and Export
Survey & Inspection Charges	Revenue & Cost	As Applicable per Shipment
GST — 18% on Indian-side Services	Tax Component	Both Import and Export

Invoice Management Features

- Auto-population of all charges directly from the job record — eliminates manual re-entry and reduces billing errors
- GST-compliant invoice generation — CGST and SGST applied for intra-state; IGST applied for inter-state transactions
- Multi-currency invoicing with exchange rate locked at the invoice generation date
- Invoice approval workflow — Finance Executive submits; Branch Manager approves before customer dispatch
- Professional PDF invoice generation with digital signature support and company branding
- Email dispatch of the invoice PDF directly from the system with delivery confirmation
- Credit note generation for billing disputes, charge adjustments, and invoice reversals
- Payment recording against invoices — supports cash, NEFT, RTGS, and cheque with full reference tracking

Accounts Receivable & Payable

Feature	Description
Customer Outstanding Dashboard	Aging analysis displayed in buckets — 0–30, 31–60, 61–90, and 90+ days overdue
Automated Payment Reminders	WhatsApp and email reminders dispatched at 7 days, 3 days, and on the due date
Vendor Payable Management	Shipping line, CFS operator, and transport vendor payment tracking and scheduling
Vendor Bill Reconciliation	Match vendor bills against job-level accrued costs for accurate P&L reporting
P&L per Shipment Job	Revenue minus cost per job; gross margin calculation with drill-down by charge type
GST Report Support	GSTR-1 and GSTR-3B summary export formatted for chartered accountant review
Bank Reconciliation	Match incoming customer payments against bank statement entries for accurate AR closure

MODULE 9 | NOTIFICATION ENGINE

The Notification Engine ensures all stakeholders receive timely, automated communications at every significant workflow milestone — eliminating the need for manual status calls and substantially improving the customer experience.

Notification Event	Channel	Recipient
Shipment Job Created	Email + WhatsApp	Customer and Operations Team

Notification Event	Channel	Recipient
Booking Confirmed with Shipping Line	Email + WhatsApp	Customer
Vessel ETD or ETA Update	WhatsApp + Mobile App Push	Customer and Operations
Document Received or Document Missing	Email	Operations Team and Customer
Bill of Entry (BOE) Filed	Email + WhatsApp	Customer and Finance Team
Customs Clearance — OOC or LEO Granted	WhatsApp	Customer and Operations
Delivery Confirmed at Consignee	WhatsApp + Email	Customer
Invoice Generated	Email with PDF Attachment	Customer
Payment Overdue — 7, 3, and 0 Days	WhatsApp + Email	Customer and Finance Team
Document Expiry Alert	Email	Operations Team
Container Detention Alert — Day 5 and Day 8	WhatsApp + Email	Operations Team and Customer
Vessel Delay Exceeding 24 Hours	WhatsApp	Customer and Operations Team

MODULE 10 | DASHBOARD & ANALYTICS

The Analytics module transforms raw operational data into actionable business intelligence — providing management with the visibility required for performance monitoring, resource allocation, and strategic decision making.

Real-Time Operational Dashboard

- Active job count segmented by type (Import and Export) and by current status stage
- Jobs requiring immediate attention — documents pending, customs examination holds, and overdue deliveries
- 14-day vessel arrival calendar for advance import planning and transport pre-allocation
- Containers at detention risk — days remaining against free-time allowance per shipping line
- Notification feed and pending approvals count segmented by approver and module

Financial Dashboard

- Month-to-date and year-to-date revenue versus set targets with variance tracking
- Outstanding receivables with full aging breakdown — visual heat map by customer and overdue band
- Top 10 customers by revenue — current month and rolling 12-month comparison
- Gross margin analysis by job type, trade lane, and customer segment
- Pending invoices list awaiting Finance approval before dispatch to customers

Management Reports

Report	Frequency	Key Metrics
Shipment Volume Report	Daily and Monthly	Jobs created, closed, and in-progress segmented by type
Revenue & Billing Report	Monthly	Invoiced amount, amount collected, and outstanding balance
Customer-wise P&L Report	Monthly and Quarterly	Revenue, cost, and gross margin per customer
Trade Lane Analysis	Monthly	Volume and revenue by origin-destination pair
Customs Performance Report	Monthly	Average clearance time and examination rate by customs station
Transport Efficiency Report	Monthly	Trips completed, delay rate, and cost per trip
Document Compliance Report	Weekly	Checklist completion rate and document rejection rate by job type

06 Operational Workflow Diagrams

ABOUT THESE WORKFLOWS

The tables below describe the complete shipment lifecycle as managed within FreightFlow ERP. Each stage corresponds to a trackable status milestone with a timestamped audit record and an automated notification trigger. Every status transition is logged with the user ID, date, time, and any associated remarks.

6.1 Export Shipment Workflow — End to End

Step	Stage	Module	Key Actions & Outputs	
	01	Customer Inquiry	CRM / Inquiry	Capture cargo details, trade lane, expected dates, and any special cargo requirements
	02	Freight Quotation	Quotation	Prepare rate sheet, apply all applicable charges, generate PDF quote, and dispatch to customer
	03	Quotation Acceptance	Quotation	Customer accepts and confirms; operations team converts approved quote into an active Export Job
	04	Job Creation	Shipment Jobs	EXP job auto-numbered; consignor and consignee details entered; job set to Active status
	05	Shipping Line Booking	Booking	Booking instructions generated from job data; BKG confirmation number recorded upon receipt
	06	Container Arrangement	Container Transport &	Empty container pickup order raised; trailer and driver assigned within the system
	07	Factory Stuffing	Container Transport &	Stuffing appointment confirmed; cargo loaded and container sealed; seal number recorded
	08	VGM Submission	Booking	Verified Gross Mass submitted to shipping line within the applicable cutoff deadline

Step	Stage	Module	Key Actions & Outputs
09	Document Collection	DMS	Invoice, packing list, COO, and all supporting export documents uploaded and verified
10	Customs Filing — Shipping Bill	Customs	SB filed on ICEGATE; RMS facilitation status or examination order recorded in system
11	Port Gate-In	Container Transport &	Truck enters port; gate-in time, vehicle weight, and VGM confirmation recorded
12	Let Export Order (LEO)	Customs	LEO obtained from customs; cargo officially authorised for loading onto the vessel
13	Vessel Departure — ATD	Booking	Actual time of departure recorded; vessel tracking activated via AIS API integration
14	BL Issuance	Booking	BL draft reviewed and approved; Original BL or e-BL issued, uploaded, and linked in DMS
15	Invoice Generation	Billing	All job charges consolidated; GST-compliant PDF invoice generated and emailed to customer
16	Job Closure	Shipment Jobs	All documents confirmed; payment received and recorded; job officially closed with closure note

6.2 Import Shipment Workflow — End to End

Step	Stage	Module	Key Actions & Outputs
01	Shipment Pre-Alert	Shipment Jobs	Pre-alert received from overseas shipper or agent; IMP job created and activated in system
02	ETA Confirmation	Booking	Vessel name, voyage number, and ETA at POD recorded; customer notified automatically via WhatsApp
03	Document Collection	DMS	Invoice, packing list, BL, and COO collected from shipper; uploaded and verified in DMS
04	BL Verification	DMS	BL details cross-checked against actual shipment particulars; amendments raised with shipper if required
05	DO Collection	Booking	Delivery Order obtained from shipping line following BL surrender or telex release confirmation
06	BOE Filing	Customs	Bill of Entry filed on ICEGATE; assessment initiated by the assigned customs officer
07	Duty Calculation & Payment	Customs / Billing	Duty computed using HS Code tariff; payment challan generated; duty payment confirmed and recorded
08	Customs Examination	Customs	Examination order updated — No Examination / First Check / Second Check with officer remarks
0	Out of Charge (OOC)	Customs	OOC granted by customs authority; cargo officially released; OOC date recorded

Step	Stage	Module	Key Actions & Outputs	
9			in job timeline	
10	CFS & Port Release	Container Transport	&	Container released from CFS or port terminal upon DO presentation; release event recorded
11	Delivery Arrangement	Container Transport	&	Trailer and driver allocated; delivery appointment communicated to consignee and driver via app
12	Delivery Confirmation	Container Transport	&	Delivery confirmed; timestamp, GPS coordinates, and receiver signature captured via mobile app
13	Empty Container Return	Container Transport	&	Empty container returned to shipping line depot; return receipt uploaded in DMS
14	Invoice Generation	Billing		All import charges invoiced; duty pass-through included if applicable; PDF sent to customer
15	Job Closure	Shipment Jobs		All clearance and delivery documents filed; payment confirmed; job officially closed

07 Database Design Overview

DATABASE PLATFORM

FreightFlow ERP uses PostgreSQL 16 as its primary relational database — selected for ACID compliance, native JSONB support, row-level security (RLS), table partitioning, and enterprise-grade performance under high concurrent load.

7.1 Core Database Tables

Table Name	Primary Key	Key Fields	Relationships
customers	customer_id (UUID)	name, gst_number, iec_code, credit_limit, payment_terms, status	→ inquiries, jobs, invoices
inquiries	inquiry_id (UUID)	customer_id, inquiry_date, shipment_type, mode, origin, destination, status	→ quotations
quotations	quotation_id (UUID)	inquiry_id, customer_id, validity_date, total_amount, status, version	→ jobs, quotation_items
quotation_items	item_id (UUID)	quotation_id, charge_type, currency, amount, quantity	→ quotations
shipment_jobs	job_id (UUID)	job_number, job_type, customer_id, quotation_id, pol, pod, incoterms, status	→ all operational tables
job_status_log	log_id (UUID)	job_id, status, changed_by, changed_at, remarks	→ shipment_jobs
bookings	booking_id (UUID)	job_id, shipping_line, bkg_number, vessel_name, voyage, etd, eta, bl_number	→ shipment_jobs

Table Name	Primary Key	Key Fields	Relationships
containers	container_id (UUID)	job_id, container_number, container_type, seal_number, status, location	→ shipment_jobs
transport_trips	trip_id (UUID)	container_id, trip_type, vehicle_number, driver_id, scheduled_dt, actual_dt	→ containers
drivers	driver_id (UUID)	name, mobile, license_number, vendor_id, status	→ transport_trips
customs_boe	boe_id (UUID)	job_id, boe_number, filing_date, assessable_value, total_duty, ooc_date	→ shipment_jobs
customs_shipping_bill	sb_id (UUID)	job_id, sb_number, sb_type, filing_date, leo_date, drawback_amount	→ shipment_jobs
documents	doc_id (UUID)	job_id, doc_type, file_path, version, status, uploaded_by, expiry_date	→ shipment_jobs
invoices	invoice_id (UUID)	job_id, customer_id, invoice_number, date, total_amount, gst_amount, status	→ invoice_items, payments
invoice_items	item_id (UUID)	invoice_id, charge_type, quantity, rate, amount, gst_rate, gst_amount	→ invoices
payments	payment_id (UUID)	invoice_id, payment_date, amount, payment_mode, reference, recorded_by	→ invoices
vendor_bills	bill_id (UUID)	job_id, vendor_id, charge_type, amount, bill_date, payment_status	→ shipment_jobs
vendors	vendor_id (UUID)	name, vendor_type, gst_number, contact, payment_terms	→ vendor_bills, trips
notifications	notif_id (UUID)	job_id, notif_type, channel, recipient, sent_at, status, message_body	→ shipment_jobs
users	user_id (UUID)	name, email, mobile, role, branch_id, is_active, last_login	→ all tables as created_by
audit_logs	audit_id (UUID)	table_name, record_id, action, old_value, new_value, changed_by, changed_at	System-wide

7.2 Database Design Principles

- UUID primary keys across all tables — prevents enumeration attacks and fully supports distributed systems
- Soft deletes via is_deleted flag and deleted_at timestamp — no hard deletes; complete audit trail preserved
- Row-Level Security (RLS) enforced in PostgreSQL — users access only data within their authorised branch scope
- Separate database schema per tenant for future SaaS multi-tenancy without any data co-mingling
- JSONB columns for flexible custom fields — records extended without requiring schema migrations
- Full-text search indexes on customer name, job number, container number, and BL number for fast lookups
- Monthly partitioned tables for audit_logs and notifications — ensures sustained query performance at scale

08 Technology Stack

8.1 Full Technology Stack

Layer	Technology	Rationale
Frontend Framework	React.js + Next.js 14 (App Router)	Server-side rendering for SEO and performance; mature component ecosystem
UI Component Library	Ant Design Pro + Tailwind CSS	Enterprise-grade components; fully responsive; highly customisable
State Management	Redux Toolkit + React Query	Global state with server cache synchronisation and real-time data refresh
Backend API	Node.js + Express.js (TypeScript)	High concurrency; JSON-native; excellent fit for microservices architecture
Alternative Backend	Django REST Framework (Python)	Preferred if AI and ML workloads are prioritised from Phase 1 onwards
Primary Database	PostgreSQL 16	ACID compliance; JSONB support; advanced indexing; open-source and proven
Cache Layer	Redis — AWS ElastiCache	Session management; real-time data caching; background job queue handling
Message Queue	RabbitMQ / AWS SQS	Asynchronous notification processing and webhook event handling
Document Storage	AWS S3 / Azure Blob Storage	Versioned, scalable document storage with integrated CDN acceleration
Full-Text Search	Elasticsearch (Phase 2+)	Advanced search across shipment records, customers, and documents
Authentication	JWT + Refresh Token Rotation + OAuth 2.0	Secure stateless authentication; Single Sign-On ready architecture
API Gateway	AWS API Gateway / Kong	Rate limiting, request routing, monitoring, and versioning
Container Platform	Docker + Kubernetes — EKS or AKS	Consistent deployments; auto-scaling; zero-downtime rolling releases
CI / CD Pipeline	GitHub Actions + ArgoCD	Automated testing, build, and deployment pipeline
Monitoring & APM	Datadog / AWS CloudWatch	Performance monitoring, error tracking, and uptime alerting
Cloud Platform	AWS (Primary) / Azure (Secondary)	Global infrastructure; fully managed services; compliance certifications
Mobile Application	React Native	Single shared codebase for native iOS and Android applications
PDF Generation	Puppeteer / PDFKit	High-fidelity PDF generation for invoices, quotations, and reports
Email Service	AWS SES / SendGrid	Transactional email delivery with bounce and delivery tracking
WhatsApp API	Twilio / Meta Cloud API	Official WhatsApp Business API for automated milestone notifications

8.2 AWS Cloud Deployment Architecture

- VPC with public and private subnets across two Availability Zones for high availability and fault tolerance
- Application Load Balancer distributing traffic to ECS Fargate or EKS containerised application services
- RDS PostgreSQL in Multi-AZ configuration with automated daily backups and seven-day point-in-time recovery

- ElastiCache Redis cluster for session storage, application caching, and message queue support
- S3 buckets for document storage with object versioning enabled and lifecycle management policies
- CloudFront CDN for frontend static assets and globally accelerated document download access
- Route 53 for DNS management with health check-based automatic failover configuration
- AWS WAF providing web application firewall protection against the OWASP Top 10 threat categories
- AWS KMS for encryption key management with automated key rotation schedule

8.3 Microservices Architecture

Service	Responsibility	Technology
API Gateway Service	Request routing, authentication enforcement, and rate limiting	Kong / AWS API Gateway
Auth Service	JWT issuance, refresh token rotation, and RBAC enforcement	Node.js + Redis
CRM Service	Customer records, contacts, KYC, and history management	Node.js + PostgreSQL
Job Service	Shipment job CRUD, status transitions, and timeline management	Node.js + PostgreSQL
Booking Service	Vessel schedules, BL lifecycle, and ETA management	Node.js + PostgreSQL
Document Service	File upload, DMS workflows, and S3 object management	Node.js + AWS S3
Customs Service	BOE, Shipping Bill, duty calculation, and clearance tracking	Node.js + PostgreSQL
Transport Service	Container, trip, driver, and GPS tracking management	Node.js + PostgreSQL
Billing Service	Invoice generation, payment recording, and P&L calculation	Node.js + PostgreSQL
Notification Service	WhatsApp, email, and push notification dispatch	Node.js + RabbitMQ
Analytics Service	Dashboard data aggregation, reports, and KPI computation	Node.js + Redis + PostgreSQL
AI Service (Phase 3)	OCR, delay prediction, smart quotations, and chatbot	Python + FastAPI + ML Models

09 Security Architecture

9.1 Authentication & Authorisation

- JWT access tokens with 15-minute expiry paired with refresh token rotation on a 7-day sliding window
- Multi-Factor Authentication via TOTP — Google Authenticator mandatory for all privileged and admin roles
- Role-Based Access Control (RBAC) enforced at the API layer — every endpoint is role-protected without exception

- Row-Level Security in PostgreSQL — users can only access data within their authorised branch or customer scope
- IP allowlisting enforced for admin panel and all system configuration endpoints
- Automatic session invalidation triggered on password change or detection of anomalous login activity

9.2 Data Security Controls

Security Control	Implementation
Encryption at Rest	AES-256 encryption applied to RDS database storage and S3 document storage layers
Encryption in Transit	TLS 1.3 enforced for all API and browser communication — HTTPS only; no HTTP fallback
Database Credentials	AWS Secrets Manager — zero hardcoded credentials in codebase, config files, or environment variables
API Key Management	AWS KMS with automated rotation schedule; keys never exposed in application logs
Document Access Control	Pre-signed S3 URLs with 15-minute expiry — no direct S3 bucket public access
Input Validation	Server-side validation enforced on all API inputs; ORM-based parameterised SQL queries
SQL Injection Prevention	Sequelize or Prisma ORM with parameterised query enforcement across all database interactions
XSS Prevention	Content Security Policy headers enforced; strict output encoding on all rendered data
CSRF Protection	Anti-CSRF tokens required on all state-changing HTTP requests
Rate Limiting	100 requests per minute per authenticated user; 1,000 per minute per source IP address
Audit Logging	All CREATE, UPDATE, and DELETE operations logged — user identity, timestamp, and before-after values
PII Data Masking	Customer PAN, Aadhaar, and bank account details masked in UI; full access restricted to authorised roles

9.3 Regulatory Compliance

- DPDP Act 2023 — India Digital Personal Data Protection Act compliance for all PII data handling and storage
- GSTN API integration via a certified GSP (Goods and Services Tax Suvidha Provider) partner
- ICEGATE API compliance for customs BOE and Shipping Bill data exchange and status tracking
- SOC 2 Type II readiness — audit controls, access logging, and monitoring designed for enterprise certification
- Data residency in India — all data stored in AWS Mumbai (ap-south-1) region for regulatory sovereignty

10 API Integration Architecture

10.1 Third-Party System Integrations

Integration	Purpose	Protocol
ICEGATE — Indian Customs Portal	BOE and Shipping Bill filing, status tracking, and duty computation	SOAP / REST
Shipping Line APIs — Maersk, MSC,	Vessel schedule, booking confirmation, and	REST / EDI

Integration	Purpose	Protocol
CMA CGM	container event tracking	
AIS Vessel Tracking — MarineTraffic	Real-time vessel position data and dynamic ETA calculation	REST
WhatsApp Business API — Meta or Twilio	Automated shipment milestone notifications to customers and staff	REST / Webhook
GSTN — Goods and Services Tax Network	GSTIN validation and e-Invoice (IRP) reporting compliance	REST
Payment Gateways — Razorpay or PayU	Online payment collection from the customer self-service portal	REST
E-Way Bill Portal — NIC	E-Way bill generation for all road transport movements within India	REST
Container Tracking — project44 or SeaRates	Multi-carrier container event and milestone tracking	REST
Currency Exchange — Fixer.io or RBI API	Real-time foreign exchange rates for multi-currency invoicing	REST
Email Service — AWS SES or SendGrid	Transactional email delivery with open, click, and bounce tracking	REST / SMTP
SMS Gateway — AWS SNS or Twilio	SMS alerts for critical operational and payment notifications	REST
Google Maps Platform	Transport route optimisation and real-time driver GPS tracking	REST
DigiLocker API	Digital verification of IEC and GST registration certificates	REST

10.2 API Design Principles

- RESTful API design conforming to the OpenAPI 3.0 specification — fully documented via interactive Swagger UI
- API versioning via URL path — /api/v1/ and /api/v2/ — with backward compatibility guaranteed within each version
- Webhook support for inbound event streams from shipping lines and customs portal status updates
- GraphQL endpoint available for mobile application — flexible field selection reduces mobile data transfer
- Idempotency keys required on all write operations — prevents duplicate record creation from network retries
- Comprehensive request and response logging maintained for debugging, auditing, and compliance purposes

11 Mobile Application Scope

PLATFORM & TECHNOLOGY

The FreightFlow mobile application is built using React Native, delivering native iOS and Android applications from a single shared codebase. The app is designed and optimised for field operations — including camera-based document capture, GPS-enabled delivery confirmation, real-time shipment monitoring, and management dashboards.

11.1 Mobile Features by User Role

Role	Mobile Application Features
Operations Executive	View and update active job status; upload documents using the device camera; receive push notifications for key shipment milestones
Transport Coordinator	Allocate drivers and vehicles to trips; view the full trip schedule; track driver GPS location in real time; confirm gate-in and delivery events
Driver	View assigned trip details with integrated navigation; capture delivery photographs; confirm delivery using OTP verification or digital signature
Customs Executive	View BOE and Shipping Bill filing status; update examination remarks; upload customs documents directly from the device
Management	Executive summary dashboard with revenue, job counts, and critical alerts; approve pending items from mobile
Customer — Portal User	Track shipment status in real time; view and download job documents; receive automated milestone notifications; download and share invoices

11.2 Offline Capability

- Critical job data and customer contact information are cached locally to support offline access during network outages
- Document uploads are queued offline and automatically synchronised to the server when network connectivity is restored
- Delivery confirmation — including GPS coordinates, timestamp, and receiver signature — captured offline and synced on reconnection

12 AI-Powered Features — Phase 3 Roadmap

AI READINESS NOTE

AI and machine learning capabilities are introduced in Phase 3, built upon the rich operational dataset generated during Phases 1 and 2. This approach ensures that all predictive models are trained on actual company data — maximising accuracy, commercial relevance, and return on investment from the moment of deployment.

AI Feature	Description	Business Value	Technology
OCR Document Reading	Automatically extract structured data from uploaded Invoice, Packing List, and BL documents	Eliminates manual data entry; reduces extraction errors by up to 90%	Google Document AI / AWS Textract
AI Invoice Data Extraction	Auto-populate BOE and Shipping Bill fields from the uploaded commercial invoice	Reduces customs filing preparation time by 50%	NLP + Document AI
Shipment Delay Prediction	Predict shipment delays based on vessel position, port congestion index, and weather data	Enables proactive customer alerts and contingency action planning	ML — Time Series Forecasting
ETA Prediction Engine	Predict actual arrival date	Improves customer	ML — Regression Models

AI Feature	Description	Business Value	Technology
	and time 48–72 hours in advance with 85%+ accuracy	experience and enables pre-booking of transport resources	
Smart Quotation Advisor	Recommend competitive freight rates based on historical win and loss data analysis	Improves quote win rates and eliminates chronic underpricing	ML — Pricing Optimisation
AI Customer Chatbot	WhatsApp and web chatbot for shipment status inquiries, document requests, and FAQs	Reduces inbound customer service calls by up to 60%	LLM + RAG — GPT-4 or Claude
Anomaly Detection	Flag unusual charges, duplicate invoices, and suspicious billing patterns	Protects revenue and prevents fraudulent billing activities	ML — Anomaly Detection
Demand Forecasting	Predict monthly shipment volume by trade lane, season, and commodity type	Supports capacity planning and operational staffing decisions	ML — Time Series Analysis

13 Implementation Plan

DELIVERY APPROACH

FreightFlow ERP is delivered in four progressive phases. Each phase produces fully functional, production-ready software — enabling the business to realise value at the earliest possible milestone while the complete enterprise platform is built methodically and iteratively.

13.1 Development Phases

Phase 0	Discovery & Design	Requirements finalisation, UI/UX wireframes, database schema design, API contracts, and development environment setup
Phase 1	Core MVP	CRM, Quotation Management, Shipment Job Management, Document Management, Basic Billing, Authentication, RBAC, and operational web dashboard
Phase 2	Full Operations Platform	Booking & Shipping, Container & Transport, Customs Clearance, Complete Billing & Accounts, WhatsApp Notifications, Management Reports, and Mobile App for iOS and Android
Phase 3	Intelligence & Scale	AI features including OCR, delay prediction, smart quotations, and chatbot; ICEGATE and shipping line API integrations; Advanced Analytics; and Customer Self-Service Portal

13.2 Phase Deliverables Summary

Phase	Name	Deliverables
Phase 0	Discovery & Design	Finalised SRS, approved UI/UX wireframes, database ERD, API contracts, and technical architecture document
Phase 1	Core MVP	10-module web application with Auth/RBAC, operational dashboard, basic reporting, and full UAT sign-off
Phase 2	Full Operations	Complete 10-module ERP platform, mobile app for iOS and Android, WhatsApp notification engine, and management reports
Phase 3	Intelligence & Scale	Full AI feature suite, live ICEGATE and shipping line API integrations, customer self-service portal, and advanced BI dashboards

14 Scalability & Future Expansion

14.1 Technical Scalability

- Horizontal auto-scaling — Kubernetes automatically scales application pods based on CPU and memory utilisation thresholds
- PostgreSQL read replicas provisioned for reporting and analytics workloads — zero impact on transactional performance
- Event-driven notification architecture — fully decoupled from the main application to scale independently
- CloudFront CDN-accelerated document delivery — reduces origin server load for document-intensive workflows globally
- Multi-region AWS deployment — Mumbai and Singapore regions configured for disaster recovery and reduced latency
- PgBouncer database connection pooling — supports thousands of concurrent users without exhausting database connections

14.2 Business Expansion Capabilities

Expansion Area	Capability	Delivery Phase
Multi-Branch Operations	Branch-level data isolation with inter-branch job transfer and consolidated reporting	Phase 2 — Included
Multi-Company Group ERP	Separate legal entity management under a single unified platform instance	Additional Scope
SaaS or White-Label Product	Multi-tenant architecture enabling FreightFlow to be offered as a SaaS product to other forwarders	Additional Scope
Warehouse Management Module	Bonded warehouse, CFS operations, and inventory tracking and management	Additional Scope
NVOCC Operations	House BL management, cargo consolidation, and deconsolidation workflows	Additional Scope
Air Freight Full Expansion	Complete AWB management, airline bookings, and CBM or KG-based charge calculations	Additional Scope

Expansion Area	Capability	Delivery Phase
Project & OOG Cargo	Out-of-Gauge, heavy lift, and break-bulk shipment management capabilities	Additional Scope
B2B Customer Self-Service Portal	Advanced self-service portal for high-volume importer and exporter clients	Phase 3 — Included

14.3 Long-Term AI Expansion Roadmap

- Phase 3 — OCR document reading, shipment delay prediction, smart quotation advisor, and AI customer chatbot
- Phase 4 — Autonomous customs pre-filing; AI-generated BOE and Shipping Bill draft preparation
- Phase 5 — Predictive container sourcing and procurement; real-time dynamic freight pricing engine
- Phase 6 and beyond — Digital freight brokerage marketplace connecting shippers directly with carriers

15 UI/UX Design Recommendations

15.1 Core Design Principles

- Information density balanced with clarity — operations staff manage high daily volumes; interface efficiency takes priority over decorative elements
- Status-first design — every list view surfaces the most critical status indicator without requiring a click-through to drill down
- Consistent colour-coded status system — Green for On Track, Amber for Attention Required, and Red for Critical or Delayed
- Single-page workflow views — complete job details visible on one scrollable page; multi-tab navigation is avoided
- Persistent sidebar navigation with quick access to active jobs, notification feed, and global search
- Instant global search — any job number, container number, BL number, or customer name navigates directly to the record

15.2 Dashboard Design Recommendations

- Summary KPI cards at page top — Active Imports, Active Exports, Revenue Month-to-Date, and Outstanding Receivables
- Shipment pipeline Kanban view — all jobs visualised simultaneously across every status stage
- Alert feed panel — time-sorted list of items requiring immediate attention with direct inline action links
- 14-day vessel arrival forward calendar — enables advance planning of import transport and customs resources
- World map view — live container and vessel positions with clickable job detail popups

15.3 Admin Panel Features

Admin Feature	Description
User Management	Create, edit, and deactivate users; assign and modify roles; reset passwords; view full login history and active sessions

Admin Feature	Description
Role & Permission Matrix	Granular permission configuration per role, per module, and per action type (Create, Edit, View, Delete, Export)
Branch Configuration	Add and edit branch profiles; assign branch managers; configure branch-level operational settings
Master Data Management	Port master, shipping line master, charge type master, and HS Code tariff rate lookup table
Notification Templates	Edit WhatsApp message templates and email body templates for each notification event type
System Configuration	Working day calendar, supported currencies, GST rate table, and invoice number series management
API Integration Monitor	External API health status dashboard; API call log viewer; credential reset and key rotation
Audit Log Viewer	Full-text search, date-range filter, and compliance-ready export of the system-wide audit trail
Data Backup Management	Manual backup trigger; backup history log; restore point selection and point-in-time recovery

16 MVP Recommendation & Development Roadmap

MVP PHILOSOPHY

The Minimum Viable Product delivers the core value proposition — digitising job management and eliminating document chaos — while gathering real operational feedback before full platform investment. Post-MVP usage data directly determines the priority of Phase 2 feature development.

16.1 Minimum Viable Product — Module Scope

MVP Module	Included Features
Authentication & Access Control	User login, RBAC enforcement, session management, and complete system audit logging
CRM — Customer Master	Customer registration, full contact management, KYC document upload and secure storage
Inquiry Management	Freight inquiry capture, cargo details, trade lane, and real-time inquiry status tracking
Quotation Builder	Rate sheet preparation, multi-line charge entry, PDF generation, email dispatch, and approval workflow
Shipment Job Management	Import and Export job creation, status transitions, and shipment timeline view
Document Management System	Document upload, mandatory checklist enforcement, verification workflow, and customer document access
Basic Billing & Invoicing	Invoice creation, charge entry, GST-compliant PDF invoice generation, and outstanding balance view
Operational Dashboard	Active jobs, revenue MTD, pending items, document alerts, and key performance indicator cards
Email Notification Engine	Automated email alerts for all core shipment milestone events

16.2 Full Development Roadmap — Milestone Plan

Milestone	Phase	Business Value Delivered
Discovery Complete	Phase 0	Finalised requirements; approved wireframes; technology stack confirmed and documented
MVP Go-Live	Phase 1	Digital job management, quotations, document management, and basic billing operational
MVP Feedback & Refinement	Phase 1 → 2	Real-world usage improvements incorporated before Phase 2 development commences
Booking & Shipping Module	Phase 2	Complete vessel management, BL lifecycle tracking, and automated ETA alerts
Container & Transport Module	Phase 2	Driver mobile app, trip management, real-time GPS tracking, and delivery confirmation
Customs Clearance Module	Phase 2	BOE and Shipping Bill management, duty calculation, LEO tracking, and OOC alerts
Full Billing & Accounts	Phase 2	Complete AR and AP management, GST export reports, and P&L per shipment job
Mobile App — iOS & Android	Phase 2	Field staff productivity tools and management dashboard on mobile devices
WhatsApp Notification Engine	Phase 2	Fully automated proactive customer communication at every shipment milestone
Customer Self-Service Portal	Phase 3	Document access, live shipment tracking, and invoice download without agent contact
ICEGATE API Integration	Phase 3	Live BOE and Shipping Bill status pulled directly from the customs portal
AI OCR & Data Extraction	Phase 3	Automated document data extraction; significant reduction in manual data entry
Full AI Suite & Advanced BI	Phase 3	Delay prediction, smart quotation advisor, and advanced business intelligence dashboards

17 Conclusion

17.1 Project Summary

FreightFlow ERP represents a transformational investment in operational efficiency, customer experience, and data-driven management capability for freight forwarding companies. By consolidating ten functional modules — from CRM and quotation management through customs clearance, billing, and business analytics — into a single cloud-native platform, the system systematically eliminates the fragmentation, human errors, and costly delays inherent in legacy paper-based and semi-automated logistics operations.

The phased delivery approach ensures measurable business value from the first MVP milestone, with progressive capability expansion aligned to operational readiness and user adoption at each stage. The API-first, cloud-native architecture guarantees that the platform will scale cost-effectively with business growth and integrate seamlessly with the broader global logistics technology ecosystem.

17.2 Key Benefits Summary

Dimension	Expected Outcome
Operational Efficiency	60% reduction in manual data entry; 80% fewer inbound customer shipment status inquiry calls
Financial Accuracy	100% charge capture per shipment job; invoice turnaround reduced from 7 days to same-day generation
Regulatory Compliance	Audit-ready documentation trail on all records; GST-compliant billing; improved customs accuracy
Customer Experience	Real-time shipment visibility; automated proactive communication at every milestone
Management Visibility	Live dashboards replacing weekly Excel reports — business decisions driven by real-time data
Business Scalability	Platform supports 10× shipment volume growth without proportional operational team expansion
Competitive Advantage	Technology-led differentiation in a commoditised industry; premium client retention and attraction

FreightFlow ERP

Digitising Freight Forwarding — One Shipment at a Time